

ANDREW LEE

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EDUCATION

William & Mary, Williamsburg, VA

Aug 2019 – May 2025 (expected)

- Bachelor of Science in Applied Mathematics, Bachelor of Science in Data Science
- Cumulative GPA: 3.98/4.00; Major GPA: 4.00/4.00; Honors: Dean's List
- Relevant Coursework: Neural Networks & Deep Learning, Statistical Learning, Mathematical Statistics, Probability Theory, Regression Analysis, Survival Analysis, Real Analysis I & II, Linear Algebra, Databases, Data Visualization

RESEARCH EXPERIENCE

Institute for Integrative Conservation, Williamsburg, VA

Jan 2024 – present

Funded Through IIC Summer Research Award (\$4,000)

- Worked with partners in U.S. Department of the Interior agencies to leverage deep learning object detection methods and thermal imagery to efficiently count migratory birds in aerial images of the Platte River of Nebraska.
- Produced original research using computer vision techniques on large-scale thermal imageries; policymakers plan to use results as an input to improve hunting regulations.
- Led one poster presentation at an undergraduate research symposium, and regularly presented findings to wildlife biologists and ecologists.

William & Mary, Department of Mathematics, Williamsburg, VA

June 2024 – present

- Supervised by Prof. Sun of William & Mary department of mathematics, collaborated with *Tanujit Dey* of Harvard Medical School.
- Carried out data analysis and collection to better understand exposures and behavioral responses to wildfire smoke by forecasting Google search rates of ten public health associated keywords.
- Researched and implemented regression for longitudinal data via mixed models (LMM) and generalized estimating equations (GEE) with varying correlation structures.

William & Mary, Department of Mathematics, Williamsburg, VA

May 2023 – Aug 2023

Funded Through Virginia Space Grant Consortium (\$4,000)

- Supervised by Prof. Hunt of William & Mary department of mathematics to study the applications of machine learning in hypersonic vehicles.
- Experimented numerous statistical learning methods for regression including multiple linear regression, random forest regressor, and support vector machines.
- Designed and trained a deep neural network architecture for predicting the Shock Train Leading Edge (STLE) of hypersonic vehicles – a direct interest of the NASA Langley Research Center.

PROFESSIONAL EXPERIENCE

Council on International Educational Exchange (CIEE), Seoul, South Korea

May 2022 – Aug 2022

Program Assistant

- Built study abroad programs with an emphasis on experiential learning for U.S. high school students and university students traveling to South Korea.
- Maintained 5+ independent program itineraries along with 400+ students' information utilizing Microsoft Office.
- Assisted communications between on-site Academic Coordinators and U.S. based CIEE staff.

Republic of Korea Army, 52nd Infantry Division, Gwangmyeong, South Korea

Sep 2020 – Mar 2022

Sergeant

- Trained, disciplined and taught different military skills; served as a squad leader for a team of seven.
- Applied principles of accounting to report financial statements monthly and responsible for creation of physical copies.
- Worked detached at Incheon International Airport as a translator during the COVID-19 pandemic to increase the efficiency of airport traffic.

RESEARCH PUBLICATIONS

1. Pickens, B., **Lee, A.**, Hunt, G., Luz-Ricca, E., Wilkerson, G., Pearse, A., Garoutte, E., Rose, R., Ostrom, B., Caven, A., & Koneff, M. Applying Artificial Intelligence to Monitor the Mid-Continent Population of Sandhill Crane (*Antigone canadensis*) at their Critical Stopover in the Platte River Valley (manuscript draft in progress).

TECHNICAL SKILLS AND INTERESTS

Technical Skills: Python, R, SQL, Linux/Bash, LaTeX, Tableau

Languages: English (native), Korean (native), Japanese (elementary)

Interests: Teaching, Investment, Cinema, Classical/Electric Guitar, Baseball